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tutorialverse: Interoperability and quality infrastructure for R tutorials

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Executive summary

Instructors and learners choosing an R tutorial often have to inspect its repository to find prerequisites and evidence that the exercises still work. Descriptions differ across formats and may leave maintenance status unclear.

tutorialverse will help instructors and learners find R tutorials suited to their needs and see the reported prerequisites, check results, and review status. Authors will describe tutorials with shared metadata. An R package will use adapters to run supported checks through existing engines and produce common reports. A searchable registry will combine descriptions, reports, and human reviews, linking readers to original lessons. Community review and scheduled checks will help keep entries current.

A [metadata-only prototype](#) already reads YAML records and reports validation failures. It does not yet render tutorials or run exercises.

Over 12 months, we will develop support for learnr, learnr2, and Quarto Live, index 5–10 existing tutorials, and use four small examples to test the adapters across teaching contexts. Carpentries-style materials will also be indexed, with stated limits on automated checks. Existing tools will provide browser execution and static hosting. We request USD 10,000 for this work.

Signatories and community support

Shaurita D. Hutchins will lead the project in collaboration with Samuel Bharti. Both are PhD candidates at the University of Alabama at Birmingham.

Hutchins will lead educational design and community development. She will develop the metadata schema, coordinate example tutorials, and establish accessibility and contribution guidance. A computational biologist trained in genetics, genomics, and bioinformatics, she brings experience in Carpentries instruction and reproducible R/Bioconductor workflows. She received a grant to develop [Prioritizing Candidate Genetic Variants Using GTEx, HuBMAP, and IDG](#) for the Common Fund Data Ecosystem (CFDE) Training Center. Built with Quarto Live, the module combines coding activities that run in the browser with knowledge checks, giving her direct experience developing interactive learning materials.

Bharti will lead software architecture and engine integration. He will build adapters that use existing engine checks and produce validation reports in a common format. A former Posit intern, he brings experience developing Shiny applications and R packages, including bioclients, biohttp, and biobouncer. His teaching work includes [r-for-beginners](#), an adaptation of Software Carpentry lessons with accompanying slides. He also maintains automated workflows for publishing package documentation through GitHub Actions and GitHub Pages. This combination of application development, teaching, and package maintenance will inform the tools and documentation that tutorial authors use.

We will jointly design the package API and registry, and share responsibility for governance and the example tutorials.

We requested public feedback through [r/Stats](#) and the [learnr webR discussion](#). Community contributor [@coatless](#) pointed us to [learnr2](#) and Quarto Live. That feedback helped us focus on shared metadata and checks, though the proposed schema still needs review. We will seek engine maintainers' feedback on the schema and the checks each adapter can support.

We will also consult Carpentries and Bioconductor educators, along with university and community instructors. We will add documented feedback and letters of support as received.

Problem

Learners need tutorials that match their goals and current knowledge; instructors need lessons they can assess and reuse in their teaching. Both need clear information about content, prerequisites, difficulty, and duration. They also need evidence that the exercises still work, since a tutorial may remain online after they break or its author stops maintaining it.

The tools used to build a lesson also affect how it runs. [learnr](#) runs tutorials as Shiny applications ([Aden-Buie et al. 2025](#)). [learnr2](#) builds on Quarto Live, which uses webR to execute R in the browser ([learnr2](#), n.d.; [Stagg 2026](#)). [tutorialverse](#) will give instructors and learners a common way to compare tutorial descriptions and check results across these engines.

Evidence of need and prior work

For his Quarto exercises, [Handel \(2025\)](#) compared [learnr](#), [webexercises](#), [checkdown](#), [exams](#), [quarto-webr](#), and [quarto-live](#). He assessed hosting and maintenance requirements alongside the supported exercise types. [Kranz's 2019 comparison](#) of RTutor and [learnr](#) shows that choices about exercise structure, teaching environments, and long-term support predate webR.

Discussions in the [learnr](#) community document difficulties [integrating custom Shiny questions](#) and a report of [Python exercises working locally but failing after deployment](#). These reports illustrate support needs; they do not establish how often tutorials fail.

The [rstats-ed project](#) addressed discovery with a user-curated list of R courses and learning materials, while acknowledging that it could not be comprehensive. [Glitr.org](#) helps users find and compare teaching repositories; its 2024 paper explicitly excludes evaluation of their educational quality ([Geest et al. 2024](#)).

[TeSS](#) aggregates training metadata, and [Bioschemas](#) provides shared descriptions for training resources ([Garcia et al. 2020](#); [Castro et al. 2023](#)). The Galaxy Training Network combines automated checks with community review and ongoing maintenance ([Hiltemann et al. 2023](#)). The [Journal of Open Source Education \(JOSE\)](#) provides formal peer review for reusable educational materials and software; [tutorialverse](#) can record that review provenance.

[tutorialverse](#) will connect these functions across R tutorial engines, mapping descriptions to existing standards and linking check results with human reviews and maintenance information. Reports will name tested exercises and dependencies and record the environment and date. Existing engines will run tutorials. New engines, browser runtimes, and learning-management systems are outside scope; static hosting remains a deployment mode to describe and check.

Proposal

The five milestones follow a tutorial from description through checking, discovery, and review.

Milestone 1: A minimal metadata schema

We will begin with eight required fields: identifier, title, source URL, lesson URL, engine or format, maintainer, license, and learning objectives. We will map these fields to Schema.org and the Bioschemas TrainingMaterial profile where applicable. Topic, level, prerequisites, packages, deployment mode, duration, accessibility information, and software versions will be optional in the initial schema. Optional review references and DOIs will identify external reviews and publications. The checking tools will generate validation dates and results.

Authors will reuse metadata from their tutorial sources and supply missing fields in a small accompanying file, without rewriting lessons. Missing values will be explicit. An adapter can run a check only when its required fields are supplied or reliably inferred. Nominations with unknown maintainers or licenses will remain pending review.

We will align the prototype with this draft and invite engine maintainers and educators to review it. Their feedback and the pilot’s measures of author effort will guide revisions before we publish version 1.0 with example records and adoption guidance.

Milestone 2: Adapters for existing engines

Adapters will connect existing engines to tutorialverse by translating their metadata and check results into a shared format. Proposed interfaces include `learnr_adapter()`, `learnr2_adapter()`, and `quarto_adapter()`, with Quarto Live support in the Quarto adapter. We will refine these interfaces with maintainers and document supported versions and features.

Adapters may invoke an engine’s existing build or checking tools; the engine remains responsible for execution. Conversion between engines is outside the adapters’ scope. A capability table will show each adapter’s support for metadata import, rendering checks, code execution, and accessibility checks. Engines can support different checks while reporting results in the same format. Carpentries-style materials can be indexed after metadata and link checks. Further checks will depend on the lesson’s build system, with those limits stated in the registry.

Milestone 3: Validation across engines

`check_tutorial()` will combine common checks with the selected adapter. It will check metadata, links, and required files, then assess declared dependencies and package versions. When supported by the engine, the adapter will test rendering and code execution, and identify missing or duplicate exercise identifiers. Basic automated accessibility checks will accompany manual keyboard checks. Reports will distinguish observed results from declared accessibility information.

Reports will identify failing fields or exercises and record the test environment, source version, and date. For every engine, they will distinguish unsupported or unrun checks from passes and failures. Interactive exercises must be tested directly: a reachable page, a successful render, or precomputed output is not enough. Automated checks alone will not establish teaching quality or full accessibility.

We will publish a reusable GitHub Actions workflow and integration guidance so authors can run checks in their own repositories without joining the registry. We will test the package on Linux, macOS, and Windows, using examples with known faults to verify that the checks detect problems.

Milestone 4: A searchable tutorial registry

The registry will link tutorial descriptions to the original repositories and published lessons. Authors will retain responsibility for content and hosting. Automated builds will use YAML entries to create the searchable website and publish `index.json`. A proposed `tutorial_search()` function will return results as an R data frame.

Readers will filter by topic, package, level, learning objective, engine, and deployment mode when those fields are available. Each entry will show duration and maintenance status when known, and distinguish metadata validity from execution checks and human review. Scheduled checks will flag broken links and outdated validation records.

The indexing pilot will cover 5–10 existing tutorials across the supported formats. We will record missing fields and manual corrections to measure author effort. Completion will require valid records and working search results; claims about execution will depend on separate adapter tests.

Milestone 5: Review, governance, and example tutorials

Authors will submit entries through structured pull requests; others can nominate tutorials. Automated checks will precede registry review of metadata completeness, scope, licensing, maintenance information, and declared accessibility characteristics. Review procedures will draw on open lesson-review practices ([The Carpentries 2022](#)). Where tutorials have undergone external pedagogical or scholarly review, we will link to that review rather than repeat it. Our guidance will distinguish nominations from author submissions and explain procedures for correcting, removing, or archiving entries. It will also specify responsibility for maintaining each part of tutorialverse.

We will use four small example tutorials to test the adapters: `renv`, package development with `usethis` and `devtools`, an introductory DESeq2 workflow, and a Carpentries R companion. We will choose representa-

tive exercises and adapt existing material under compatible licenses. Bharti’s r-for-beginners lessons and Bioconductor training resources offer starting points.

These examples will test metadata mapping, validation reports, and each engine’s checks; the indexing pilot will test how existing tutorials are described and found. The four examples will collectively cover all three adapters, without requiring every tutorial to run under every engine. Tests will include browser interaction where supported and local or server execution where needed. Three or four small integration templates will show authors how to connect existing lessons through their chosen engine.

Project plan

Preliminary work

The [tutorialverse prototype](#) reads YAML metadata and checks required fields, field types, engine identifiers, URL syntax, and duplicate IDs when other record IDs are supplied. It returns structured pass/fail results and field-level issues. Local tests pass using synthetic records, including records labeled learnr, learnr2, and Quarto Live. The prototype does not contact lesson URLs, render tutorials, or run exercises. Engine integration and indexing real tutorials remain proposed work. Our review of existing tools and [community feedback](#) informed this scope.

Schedule and working practices

Period	Work and completion criteria
Months 1–2	Review the eight-field schema, map metadata from existing tutorials into pilot records, establish repositories and governance, and begin the example tutorials.
Months 3–5	Implement the three adapters and <code>check_tutorial()</code> . Publish a capability table and common report format. Test valid examples and known failures.
Months 6–8	Build registry ingestion and search. Publish <code>index.json</code> and the R search interface. Run the pilot records through the package and document author effort.
Months 9–10	Pilot nominations and human review with external contributors. Test stale-entry handling and gather usability feedback.
Months 11–12	Complete example tutorial and integration-template checks. Address high-priority feedback and release version 1.0 of the package, specification, and registry with maintenance guidance.

We will work openly on GitHub, using issues and milestones to track progress and pull requests for code review. We will publish monthly updates and milestone reports and invite community feedback. Package tests will use testthat, with renv to reproduce software environments. We will publish documentation with pkgdown and distribute the package through R-universe where appropriate.

Code will use the MIT license. Newly authored content and documentation will use CC BY 4.0 where compatible with source obligations; adaptations will retain required attribution and licensing. We will share release announcements and integration guidance with R education communities.

Budget and funding

Requested budget: USD 10,000. This funds 200 hours at USD 50/hour, allocated 60% to project lead Hutchins (120 hours; USD 6,000) and 40% to Bharti (80 hours; USD 4,000). No hardware, cloud, travel, or fee costs are included.

Milestone	Hutchins (h)	Bharti (h)	Total (h)	USD
Minimal schema	18	12	30	1,500
Adapters	30	20	50	2,500
Validation package	33	22	55	2,750
Registry	21	14	35	1,750
Review, governance, and examples	18	12	30	1,500
Total	120	80	200	10,000

Milestone	Hutchins (h)	Bharti (h)	Total (h)	USD
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If funding is reduced, the schema and validation tools can be used independently of the registry.

Risks and maintenance

Adoption is the main uncertainty. Authors may find the schema burdensome if it duplicates metadata they already maintain. We will reuse existing metadata and use early consultation and the indexing pilot to measure author effort, then revise the requirements before finalizing version 1.0.

Engine changes could break adapters or expose differences in test coverage. We will document supported versions and browser or server requirements, and use the example tutorials to detect regressions.

We will document contribution and governance procedures so others can maintain the registry without relying on us for every change. Entries will name maintainers and retain source links, validation dates, and status. Scheduled checks and an archival policy will help contributors address stale resources.

Success

Learners will be able to judge whether a tutorial suits their goals, and instructors whether they can reuse it. The registry will show available requirements, check results, review status, and missing information. Authors can also describe and check tutorials in their own repositories without joining the registry.

The table compares documented results with grant-end targets. We will update the baseline at grant start to include any completed pilot work.

Measure	Documented baseline	Grant-end target
Adapters with capability reports	0 demonstrated	3 initial adapters
Tutorials indexed through tutorialverse	0 demonstrated	5–10 across supported formats
External reviewers of schema, adapters, or tutorials	0 documented reviews	At least 3
Example tutorials tested in CI	0 demonstrated	4, collectively covering all adapters
Integration templates	0 demonstrated	3–4
Reusable tutorial-validation workflow	0 demonstrated	At least 1

Completion requires version 1.0 of the schema, R package, and registry. The registry pilot must also test published review and governance policies, including procedures for nominations, human review, and handling stale resources.

Validation will apply only to the checks reported for each engine and environment. During the grant, we will track external submissions and review participation, and measure the effort authors spend supplying missing metadata. After release, we will use outdated check results and broken links to prioritize maintenance, and package coverage to identify gaps in the registry.

Future work could add adapters for other tutorial engines or connect the registry to pkgdown, R-universe, and educational catalogs. Multilingual metadata, richer accessibility checks, and instructor collections remain outside the initial funded scope.

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